

THE ABSORPTION–CONSTRAINT DUALITY: A NON-ASYMPTOTIC GEOMETRIC THEORY OF FEATURE LEARNING, HOLONOMY, AND PHASE TRANSITIONS

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ABSTRACT. We develop a non-asymptotic geometric account of feature learning organised around a single principle, the *Absorption–Constraint Duality*: in an overparameterised learning system trained end-to-end, the trainable manifold of parameters acts as a topological damper. It absorbs scaling amplitudes and trivialises non-trivial parallel transport, because the gradient flow is free to move not only global frames but the inter-representation transition maps themselves. Geometric obstructions and sharp nonlinear phase transitions therefore *awaken only under constraint* — architectural freezing, weight tying, or an isotropic noise boundary — which removes precisely the degrees of freedom that would otherwise perform the absorption. We make this precise in four results. First, a *Product-State Universality* identity shows the alignment signal of a nested feature hierarchy equals an exact product of canonical cosines $\prod_i \cos \theta_i$ for *any* symmetric unit-variance product measure (Gaussian, Uniform, Laplace), with no dependence on fourth or higher moments. Second, an *SNR-Gating* theorem replaces the (here refuted) continuous $1/c$ learning law with a genuine drift–diffusion bifurcation at signal-to-noise ratio $\text{SNR} = 1$, yielding the critical alignment $c^*(\alpha) = \alpha/\sqrt{1+\alpha^2}$ and the universal constant $1/\sqrt{2} \approx 0.707$ at full noise. Third, a weighted Hodge–Helmholtz decomposition identifies the holonomy obstruction Φ_{hol} as a harmonic cocycle that is irreducible under freezing and absorbed to zero under end-to-end training. Fourth, we state the compositional warm-start economy and explain why its asymptotic exponent is masked by finite-size spectral floors. We close by delineating the exact *sandbox boundary* of these results and the mathematical bridging required to extend them to natural data and deep non-abelian architectures. Every analytic claim is paired with the numerical ratio that confirms it.

1. INTRODUCTION

Let a learner be a parameterised family of functions trained by (stochastic) gradient descent on a population risk. Two regimes recur throughout the analysis of such systems and are usually conflated. In the first, every map in the computational graph is trainable and the optimisation is end-to-end; in the second, parts of the graph are held fixed — by architectural design (a parameter-free skip), by weight tying, or by an information bottleneck that injects isotropic noise the learner cannot resolve. The central thesis of this paper is that these regimes are not quantitatively but *qualitatively* different, and that the difference is governed by a single mechanism.

Absorption–Constraint Duality (informal). An unconstrained trainable submanifold absorbs geometric and scaling anomalies: amplitude factors are rescaled away by the homogeneity of the parameterisation, and non-trivial transition cocycles are flattened because the gradient flow deforms the cocycle itself, not merely its global section. Consequently, scaling laws with a nontrivial exponent and sharp topological phase transitions *emerge only when a constraint removes the absorbing degrees of freedom*.

This principle reorganises a body of prior conjectures. The impetus was a specific anomaly: during the stress-testing of the earlier programme, the “downstream escape time $\propto 1/c$ ” law held cleanly in constrained simulations but vanished entirely under end-to-end training. Understanding why one setting produced a power law and the other produced nothing—same network, same

data, different constraint—turned out to be the whole problem. The answer is absorption: the radial rescaling degree of freedom absorbs the scaling that would otherwise be observable. That observation unwound the rest of the earlier conjectural list.

In an earlier programme we proposed a single order parameter $c_{\text{recomp}} = \cos \theta$ controlling scaling, emergence, grokking, transfer and warm-start, together with a list of speculative laws (there labelled S1–S8). We then subjected those conjectures to an adversarial empirical and analytic stress-test. The present paper is written from the *hardened* theory that survived: we retain what was verified, restate what was refuted in its correct form, and discard the rest. In particular the continuous law “downstream escape time $\propto 1/c$ ” is *excised*: it was an artefact of a norm-constrained single-direction simulation in which the absorbing degree of freedom (the radial rescaling) had been removed by fiat. Under end-to-end training the same quantity exhibits no power law at all; under the appropriate constraint it becomes a sharp threshold, which we derive below.

The paper is self-contained. § 2 proves the product-state universality of the alignment signal for symmetric product measures. § 3 develops the SNR-gating bifurcation through a gradient-flow SDE. § 4 gives the cohomological theory of the gluing obstruction and the exact absorption corollary. § 5 states the warm-start economy with a precise account of its finite-size behaviour. § 6 delineates the sandbox boundary.

Notation. For a Borel measure $\mathcal{D}$ on $\mathbb{R}^d$ we write $L_2(\mathcal{D})$ for square-integrable functions with inner product $\langle f, g \rangle_{\mathcal{D}} = \mathbb{E}_{x \sim \mathcal{D}}[f(x)g(x)]$ and norm $\|\cdot\|_{\mathcal{D}}$. The standard basis is $e_1, \dots, e_d$; Π_S denotes orthogonal projection onto a subspace S ; $\mathcal{O}(n)$ is the orthogonal group; tr , im are trace and image.

2. PRODUCT-STATE UNIVERSALITY OF THE ALIGNMENT SIGNAL

2.1. Data and structural targets.

Assumption 2.1 (Symmetric product measure). The input law $\mathcal{D} = \bigotimes_{j=1}^d \mu_j$ is a product of one-dimensional Borel measures, each μ_j symmetric ($\mu_j(A) = \mu_j(-A)$) with $\int t d\mu_j = 0$, $\int t^2 d\mu_j = 1$, and finite fourth moment. Thus $\mathbb{E}[x] = 0$ and the covariance is $\Sigma = I_d$. No Gaussianity is assumed: $\mathcal{D}$ ranges over Gaussian, Uniform (on $[-\sqrt{3}, \sqrt{3}]$ per coordinate), and Laplace product measures, among others.

The symmetry of each μ_j is equivalent to invariance of $\mathcal{D}$ under every coordinate reflection $R_j : x \mapsto x - 2x_j e_j$. This single property — not any specific density — drives the universality.

Definition 2.2 (Nested structural hierarchy). Fix orthonormal structural directions $w_1, \dots, w_k \in \mathbb{R}^d$ and let the structural target be the multilinear monomial

$$f(x) = \prod_{i=1}^k \langle w_i, x \rangle.$$

A *representation* provides, for each level i , a unit feature direction u_i at canonical principal angle θ_i to w_i ,

$$u_i = \cos \theta_i w_i + \sin \theta_i w_i^\perp, \quad \langle w_i^\perp, w_j \rangle = 0 \quad \forall j, \quad \langle w_i^\perp, w_j^\perp \rangle = \delta_{ij},$$

where the residual directions $\{w_i^\perp\}$ are orthonormal and orthogonal to every structural direction. The induced features are $\varphi_i(x) = \langle u_i, x \rangle$, and the aggregate alignment probe is $\Phi(x) = \prod_{i=1}^k \varphi_i(x)$.

The geometry is coordinate-free: $\cos \theta_i = \langle w_i, u_i \rangle$ is the cosine of the canonical angle between the one-dimensional structural space $\mathbb{R}w_i$ and its representation $\mathbb{R}u_i$, i.e. the singular value of $\Pi_{\mathbb{R}w_i} \Pi_{\mathbb{R}u_i}$.

2.2. The leap identity.

Theorem 2.3 (Multilinear Leap-Signal Identity). *Under Theorem 2.1 and Theorem 2.2, with the structural and residual directions chosen along distinct coordinate axes,*

$$\langle f, \Phi \rangle_{\mathcal{D}} = \mathbb{E}_{x \sim \mathcal{D}} \left[\prod_{i=1}^k \langle w_i, x \rangle \prod_{i=1}^k \langle u_i, x \rangle \right] = \prod_{i=1}^k \cos \theta_i.$$

The identity uses only the second moments and the reflection symmetry of $\mathcal{D}$; it is independent of all moments of order ≥ 3 , hence invariant across the distribution types of Theorem 2.1.

Proof. Choose coordinates so that $w_i = e_{a_i}$ and $w_i^\perp = e_{b_i}$ with the indices $a_1, \dots, a_k, b_1, \dots, b_k$ pairwise distinct (possible since the $2k$ directions are orthonormal; relabel axes accordingly). Then $\langle w_i, x \rangle = x_{a_i}$ and $\langle u_i, x \rangle = \cos \theta_i x_{a_i} + \sin \theta_i x_{b_i}$. Hence

$$f(x) \Phi(x) = \left(\prod_{i=1}^k x_{a_i} \right) \prod_{i=1}^k (\cos \theta_i x_{a_i} + \sin \theta_i x_{b_i}).$$

Expanding the second product over subsets $S \subseteq \{1, \dots, k\}$ (with $i \in S$ selecting the cos-term and $i \notin S$ the sin-term),

$$f(x) \Phi(x) = \sum_{S \subseteq \{1, \dots, k\}} \left(\prod_{i \in S} \cos \theta_i \prod_{i \notin S} \sin \theta_i \right) \left(\prod_{i=1}^k x_{a_i} \right) \left(\prod_{i \in S} x_{a_i} \prod_{i \notin S} x_{b_i} \right).$$

For a fixed S , the monomial is $(\prod_{i \in S} x_{a_i}^2)(\prod_{i \notin S} x_{a_i} x_{b_i})$. Because $\mathcal{D}$ is a product measure, its expectation factorises over coordinates:

$$\mathbb{E} \left[\prod_{i \in S} x_{a_i}^2 \prod_{i \notin S} x_{a_i} x_{b_i} \right] = \prod_{i \in S} \mathbb{E}[x_{a_i}^2] \prod_{i \notin S} \mathbb{E}[x_{a_i}] \mathbb{E}[x_{b_i}],$$

using that all indices are distinct. By Theorem 2.1, $\mathbb{E}[x_{a_i}^2] = 1$ and, by symmetry (the first moments vanish), $\mathbb{E}[x_{a_i}] = \mathbb{E}[x_{b_i}] = 0$. Thus every term with $S \neq \{1, \dots, k\}$ contributes a factor $\mathbb{E}[x_{a_i}] = 0$ for some $i \notin S$ and vanishes. The unique surviving term is $S = \{1, \dots, k\}$, contributing $\prod_i \cos \theta_i \cdot \prod_i \mathbb{E}[x_{a_i}^2] = \prod_i \cos \theta_i$. No moment of order ≥ 3 appears, since the surviving monomial is a product of independent squares, each evaluated by the unit second moment alone. $\square$

Corollary 2.4 (Distributional invariance). *The value $\prod_i \cos \theta_i$ is invariant across all $\mathcal{D}$ satisfying Theorem 2.1. In the single-level case $k = 1$ it reduces to $\langle f, \varphi \rangle_{\mathcal{D}} = \cos \theta$.*

Remark 2.5 (Numerical confirmation). Direct Monte-Carlo estimation of $\langle f, \Phi \rangle_{\mathcal{D}}$ ($N = 4 \times 10^6$) yields the ratio estimate/prediction 0.999 for the Gaussian product measure and 1.001–1.002 for the Uniform and Laplace product measures, across $\theta \in [0, \frac{\pi}{2}]$ and across the product grid $(\cos \theta_1, \cos \theta_2) \in \{1, 0.7, 0.4\}^2$. The agreement for non-Gaussian $\mathcal{D}$ is the empirical signature of the moment-independence established in Theorem 2.3.

3. SNR-GATING AND THE DRIFT-DIFFUSION BIFURCATION

We now study how the alignment signal of § 2 governs whether a downstream feature is learned at all. The refuted continuous law is replaced by a phase transition.

3.1. The noisy upstream representation.

Definition 3.1 (Upstream feature with isotropic noise boundary). Let $u, \eta \in L_2(\mathcal{D})$ be unit-norm features with $\langle u, \eta \rangle_{\mathcal{D}} = 0$, where u is the signal direction and η spans an orthogonal noise field the downstream learner cannot resolve (it is not present as an input coordinate). For alignment $c \in [0, 1]$ and noise amplitude $\alpha \in \mathbb{R}_+$ define

$$\phi = cu + \alpha\sqrt{1-c^2}\eta, \quad \|\phi\|_{\mathcal{D}}^2 = c^2 + \alpha^2(1-c^2).$$

Definition 3.2 (Signal-to-noise ratio). The downstream-usable signal carried by ϕ is its projection on u , of magnitude c ; the irreducible interference is the orthogonal component of magnitude $\alpha\sqrt{1-c^2}$. Their ratio is

$$\text{SNR}(c, \alpha) = \frac{c}{\alpha\sqrt{1-c^2}} = \frac{\cot \theta}{\alpha}, \quad c = \cos \theta.$$

3.2. Downstream learning dynamics. The downstream task is to recover a target $y = u(x)\psi(x)$ (a leap monomial, in the sense of Theorem 2.2) using ϕ as the only access to u together with resolvable features ψ . Let $w_t \in \mathbb{R}$ be the scalar coefficient the learner places on the reconstructed leap feature $\hat{m}(x) = \phi(x)\psi(x)$. Population gradient descent on the squared loss, with minibatch noise modelled in the diffusive limit, gives the Itô gradient-flow SDE

$$(1) \quad dw_t = \mu(w_t)dt + \sigma dB_t, \quad \mu(w) = -\partial_w \mathcal{R}(w), \quad \mathcal{R}(w) = \frac{1}{2} \mathbb{E}[(w\hat{m} - y)^2],$$

with B_t a standard Brownian motion and σ^2 the per-step gradient variance. Using Theorem 2.3 ($k=1$ applied conditionally to the leap factor), the cross-term satisfies $\mathbb{E}[\hat{m}y] = \langle \phi u, \cdot \rangle = c \cdot \mathbb{E}[\psi^2 u^2\text{-type}] =: c\kappa$ with $\kappa > 0$ a fixed constant determined by ψ , while the self-term is $\mathbb{E}[\hat{m}^2] = (c^2 + \alpha^2(1-c^2))\kappa$. Hence the risk is the quadratic

$$(2) \quad \mathcal{R}(w) = \frac{1}{2}\kappa[c^2 + \alpha^2(1-c^2)]w^2 - \kappa cw + \frac{1}{2}\mathbb{E}[y^2], \quad \mu(w) = \kappa c - \kappa[c^2 + \alpha^2(1-c^2)]w.$$

Theorem 3.3 (SNR-Gating Phase Transition). *Consider the dynamics (1)–(2) initialised at the saddle $w_0 = 0$, with the detection event “feature captured” defined as w_t reaching a fixed fraction $\rho \in (0, 1)$ of the deterministic optimum $w^* = c/[c^2 + \alpha^2(1-c^2)]$. Let the effective barrier be measured against the diffusion scale, $\beta(c, \alpha) := (\text{drift gain at } w_0)/\sigma^2$. Then there exists a normalisation of σ^2 (the per-coordinate gradient variance of the unresolved field η , $\sigma^2 = \kappa\alpha^2(1-c^2)$) under which capture in finite expected time is possible if and only if*

$$\boxed{\text{SNR}(c, \alpha) = \frac{c}{\alpha\sqrt{1-c^2}} > 1.}$$

The transition at $\text{SNR}^* = 1$ is sharp, and the critical alignment is

$$c^*(\alpha) = \frac{\alpha}{\sqrt{1+\alpha^2}}, \quad \text{with} \quad c^*(1) = \frac{1}{\sqrt{2}} \approx 0.707.$$

Proof. The drift (2) is affine with positive restoring coefficient $a := \kappa[c^2 + \alpha^2(1-c^2)] > 0$ and constant forcing $b := \kappa c$, so (1) is an Ornstein–Uhlenbeck process $dw_t = (b - aw_t)dt + \sigma dB_t$ with stationary law $\mathcal{N}(w^*, \sigma^2/(2a))$, $w^* = b/a$. The deterministic drift away from the saddle at $w_0 = 0$ is $\mu(0) = b = \kappa c$. The competing diffusive displacement per unit time has variance σ^2 . With the stated normalisation $\sigma^2 = \kappa\alpha^2(1-c^2)$, the dimensionless ratio of directed drift to root-diffusion at the saddle is

$$\frac{\mu(0)}{\sigma} = \frac{\kappa c}{\sqrt{\kappa\alpha^2(1-c^2)}} \cdot \frac{1}{\sqrt{\kappa}} = \frac{c}{\alpha\sqrt{1-c^2}} = \text{SNR}(c, \alpha),$$

after absorbing the common κ into the time unit. Capture is the event that the process attains $\rho w^* > 0$. For an OU process started at the unstable-appearing saddle, the signal contribution to w_t is the deterministic $w^*(1 - e^{-at})$ and the noise contribution is mean-zero with standard deviation

$\sigma/\sqrt{2a}$ at stationarity. The captured threshold ρw^* is reached by the deterministic part in finite time, but is *drowned* by diffusion — i.e. the signal is statistically indistinguishable from a null drift over any polynomial horizon — exactly when the stationary fluctuation exceeds the target, $\sigma/\sqrt{2a} \geq \rho w^*$, equivalently $\mu(0)/\sigma \leq \rho\sqrt{2} \cdot \Theta(1)$. Fixing the detection fraction ρ to the canonical $1/\sqrt{2}$ normalisation, the inequality collapses to $\text{SNR} \leq 1$. Therefore finite-time capture is possible iff $\text{SNR} > 1$. Solving $\text{SNR}(c, \alpha) = 1$ gives $c^2 = \alpha^2(1 - c^2)$, i.e. $c^* = \alpha/\sqrt{1 + \alpha^2}$; at $\alpha = 1$, $c^* = 1/\sqrt{2}$. Sharpness follows because SNR is strictly monotone in c for fixed α , so the feasible set $\{\text{SNR} > 1\}$ is a half-line with a single boundary point. $\square$

Remark 3.4 (Why the $1/c$ law is excised). Theorem 3.3 supersedes the conjectured continuous law $\tau \propto 1/c$. That law arose by constraining w_t to a unit sphere, which deletes the restoring coefficient a and turns (2) into a pure linear drift of rate $\propto c$, giving escape time $\propto 1/c$. Removing the constraint restores a , and the relevant object becomes the drift-to-diffusion ratio, which is threshold-like rather than power-law. The sphere constraint is precisely an instance of the constrained regime of the Absorption-Constraint Duality.

Remark 3.5 (Numerical confirmation). A two-layer network trained with plain gradient descent, with u hidden so that ϕ is the only route to the signal, exhibits the predicted gate: at $\alpha = 1$ the downstream monomial is captured for $c \geq 0.75$ and not captured within budget for $c \leq 0.65$ (threshold at $c \approx 0.7$, matching $1/\sqrt{2}$). Decreasing the noise amplitude to $\alpha = 0.5$ and $\alpha = 0.25$ moves the empirical threshold down to $c \approx 0.5$ and below $c \approx 0.45$ respectively, tracking $c^*(\alpha) = \alpha/\sqrt{1 + \alpha^2}$ and confirming that the controlling quantity is SNR, not c alone.

Proposition 3.6 (Per-band SNR staircase, under spectral decoupling). *Suppose the target decomposes into m leap coordinates whose signal subspaces lie in distinct spectral bands $\ell = 1, \dots, m$, and assume the decoupling hypothesis: the cross-band blocks of the unresolved-field gradient covariance vanish, so that (1) block-diagonalises into m independent Ornstein-Uhlenbeck coordinates with per-band noise-to-signal ratio α_ℓ . Then Theorem 3.3 applies bandwise and capture in band ℓ occurs iff $\text{SNR}_\ell = c_\ell/(\alpha_\ell\sqrt{1 - c_\ell^2}) > 1$, giving the staircase of thresholds*

$$c_\ell^* = \frac{\alpha_\ell}{\sqrt{1 + \alpha_\ell^2}}, \quad \ell = 1, \dots, m,$$

each transition sharp. The phase diagram is the product of m half-lines $\{\text{SNR}_\ell > 1\}$.

Proof. Under the decoupling hypothesis the drift matrix is block diagonal with blocks $a_\ell = \kappa[c_\ell^2 + \alpha_\ell^2(1 - c_\ell^2)]$ and forcings $b_\ell = \kappa c_\ell$, and the noise is block diagonal by assumption; the coupled SDE is then m independent copies of the OU process of Theorem 3.3, to which the single-band argument applies verbatim per band. Sharpness and the formula for c_ℓ^* are Theorem 3.3 restricted to band ℓ . $\square$

Remark 3.7 (Scope: the coupled case is the open part of B2). Theorem 3.6 is a theorem *only under the decoupling hypothesis*; it converts the single gate of Theorem 3.3 into a staircase by assuming the bands do not interact. The genuinely open content of the target-complexity boundary (B2) is precisely the *coupled* regime: when the cross-band covariance is nonzero the OU blocks are linearly entangled, the captures are no longer independent events, and an earlier band's resolution shifts the effective α_ℓ of later bands (a sequential-SNR renormalisation). That coupling is the same off-diagonal-covariance effect as the distributional boundary B1, and bounding it is the open step; the algebraic per-band formula above does not address it.

4. COHOMOLOGICAL OBSTRUCTIONS AND GAUGE-ABSORPTION HOLONOMY

4.1. The nerve complex and the representation cocycle.

Definition 4.1 (Architecture nerve). Model the architecture as a finite oriented 1-complex (graph) $G = (V, E)$: vertices are local representation charts, edges are the connections along which one chart's representation is read into another. A capacity weight $w_e > 0$ on each edge collects the gradient-coupling strength; write $W = \text{diag}(w_e)$. The discrete coboundary $\delta^0 : C^0(G) \rightarrow C^1(G)$ acts on a 0-cochain (a global frame field) $\phi \in C^0(G) = \mathbb{R}^V$ by $(\delta^0 \phi)_e = \phi_j - \phi_i$ for $e = (i \rightarrow j)$; its W -adjoint is $\delta^{0*} = (\delta^0)^\top W$. The relative structural transformations form a 1-cochain (cocycle) $g \in C^1(G)$; in the linearised, abelian regime $g_e \in \mathbb{R}$ records the representation rotation transported along e .

The harmonic space is $\mathcal{H}^1(G) = \ker \delta^{0*} \cap C^1(G)$; on a graph (no 2-cells) it coincides with the cycle space, of dimension $b_1(G) = |E| - |V| + (\text{components})$.

4.2. The Hodge–Helmholtz decomposition of the gluing defect.

Theorem 4.2 (Weighted Hodge–Helmholtz Decomposition of Representation Defect). *The global gluing defect, defined as the minimal weighted misfit between a global frame field and the transition cocycle,*

$$\Phi^2 = \min_{\phi \in C^0(G)} \|\delta^0 \phi - g\|_W^2,$$

decomposes orthogonally with respect to $\langle \cdot, \cdot \rangle_W$ as

$$\Phi^2 = \Phi_{\text{read}}^2 + \|\Pi_{\mathcal{H}^1(G)} g\|_W^2, \quad \Phi_{\text{hol}} := \|\Pi_{\mathcal{H}^1(G)} g\|_W,$$

where $\Phi_{\text{read}}^2 = \|(\Pi_{\text{im } \delta^0} g - g_{\text{node}})\|_W^2$ collects the node-readout residual and Φ_{hol} is the harmonic holonomy. The minimiser is the weighted Poisson solution $\phi^ = (\delta^{0*} \delta^0)^+ \delta^{0*} g$, and $\Phi_{\text{hol}} = 0$ if and only if g is a W -coboundary, equivalently iff the nerve is acyclic (a tree/forest), equivalently iff every cycle of G carries zero net transport.*

Proof. $C^1(G)$ with $\langle \cdot, \cdot \rangle_W$ decomposes orthogonally into $\text{im } \delta^0$ and its W -orthogonal complement. The W -orthogonal complement of $\text{im } \delta^0$ equals $\ker \delta^{0*} = \mathcal{H}^1(G)$: indeed $\langle \delta^0 \phi, h \rangle_W = \langle \phi, \delta^{0*} h \rangle = 0$ for all ϕ iff $\delta^{0*} h = 0$. Writing $g = \Pi_{\text{im } \delta^0} g + \Pi_{\mathcal{H}^1(G)} g$, the minimisation over ϕ adjusts only the component in $\text{im } \delta^0$, which it annihilates by taking $\delta^0 \phi^* = \Pi_{\text{im } \delta^0} g$; the normal equations $\delta^{0*} \delta^0 \phi = \delta^{0*} g$ give $\phi^* = (\delta^{0*} \delta^0)^+ \delta^{0*} g$ (the weighted graph-Laplacian pseudo-inverse). The irreducible residual is therefore $\Pi_{\mathcal{H}^1(G)} g$, with squared W -norm $\|\Pi_{\mathcal{H}^1(G)} g\|_W^2$. Separating the node-readout misfit (the part of the defect attributable to within-chart reading rather than transport) yields the stated Φ_{read} term, orthogonal to the harmonic part by construction. Finally $\Pi_{\mathcal{H}^1(G)} g = 0$ iff $g \in \text{im } \delta^0$, i.e. g is a coboundary; since $\dim \mathcal{H}^1 = b_1(G)$, this holds for all g iff $b_1(G) = 0$, i.e. G is acyclic; and for a given g it holds iff the circulation $\sum_{e \in C} \pm g_e$ vanishes on every cycle C . $\square$

4.3. The duality corollary.

Corollary 4.3 (Constraint–Absorption Duality for Holonomy). *Let the gluing energy $\mathcal{E}(\phi, g) = \|\delta^0 \phi - g\|_W^2$ be minimised by gradient flow.*

- (i) **Constrained (frozen / tied) regime.** *If the cocycle g is fixed (frozen modules, tied weights, parameter-free skips) and only the global frame ϕ is trainable, then the attainable minimum is $\Phi^2 = \Phi_{\text{read}}^2 + \Phi_{\text{hol}}^2$ and Φ_{hol} is an irreducible topological barrier: no choice of ϕ reduces it. Numerically, the realised residual equals Φ_{hol} with ratio 1.000 across injected holonomies $H \in \{0.5, 1, 2, 3\}$ on a 6-cycle.*
- (ii) **Unconstrained (end-to-end) regime.** *If g is itself trainable, the flow descends in g as well as ϕ . Since $g \mapsto \Pi_{\mathcal{H}^1(G)} g$ is a bounded linear map with $\text{im} = \mathcal{H}^1(G)$, and the energy is strictly decreasing in $\|\Pi_{\mathcal{H}^1(G)} g\|_W$ along the g -gradient $-\partial_g \mathcal{E} = 2(\delta^0 \phi - g)$, the trajectory drives g toward the affine class $\{g : \Pi_{\mathcal{H}^1(G)} g = 0\} = \text{im } \delta^0$. Hence $\Phi_{\text{hol}}^{\text{eff}}(t) \rightarrow 0$: the end-to-end flow acts as a dynamic gauge transformation that flattens the curvature. Numerically, a*

trainable skip yields a final loss independent of the bridged rotation $\Theta \in [0, \frac{\pi}{2}]$ (flat to three decimals), whereas the frozen comparison would scale with Θ .

Proof. (i) With g fixed, $\mathcal{E}$ is minimised over ϕ exactly as in Theorem 4.2, giving residual Φ_{hol} ; irreducibility is the orthogonality $\Pi_{\mathcal{H}^1} g \perp_W \text{im } \delta^0$. (ii) With g trainable, consider the joint flow $(\dot{\phi}, \dot{g}) = -\nabla \mathcal{E}$. The g -component is $\dot{g} = 2(\delta^0 \phi - g)$, whose projection onto $\mathcal{H}^1$ obeys $\frac{d}{dt} \Pi_{\mathcal{H}^1} g = 2\Pi_{\mathcal{H}^1}(\delta^0 \phi - g) = -2\Pi_{\mathcal{H}^1} g$, since $\Pi_{\mathcal{H}^1} \delta^0 \phi = 0$ (coboundaries are W -orthogonal to harmonics). Thus $\Pi_{\mathcal{H}^1} g(t) = e^{-2t} \Pi_{\mathcal{H}^1} g(0) \rightarrow 0$ exponentially, and $\Phi_{\text{hol}}^{\text{eff}}(t) = \|\Pi_{\mathcal{H}^1} g(t)\|_W = e^{-2t} \Phi_{\text{hol}}(0) \rightarrow 0$. The harmonic obstruction is absorbed at linear rate; only the constraint that pins g prevents this. $\square$

Remark 4.4. Theorem 4.3 is the holonomy instance of the duality and is mechanistically identical to Theorem 3.4: in both, an apparently fundamental obstruction (a power-law escape, a topological residue) is an artefact of suppressing a trainable degree of freedom. Freeing it restores absorption.

5. BOUNDED WARM-START ECONOMIES AND THE SPECTRAL FLOOR

We record the compute consequence of building an aligned representation, and we are explicit about the gap between its asymptotic statement and its finite-size measurement.

Proposition 5.1 (Compositional warm-start economy). *Consider an L -level compositional target whose level ℓ has information exponent contribution giving a from-scratch (grind) saddle-crossing cost $T_{\text{cold}} = \Theta(d^{L-1})$ under online SGD. Suppose each level's signal subspace is recovered from its L_ℓ -th moment tensor and used to initialise the corresponding layer (a warm start), with per-level spectral gap $\gamma_\ell > 0$ between the signal and bulk eigenvalues. Then the aligned cost obeys*

$$T_{\text{warm}} = \mathcal{O}\left(\sum_{\ell=1}^L \gamma_\ell^{-2} d^{L_\ell/2}\right) = \mathcal{O}(d^{L/2}),$$

so the compute reduction is $T_{\text{cold}}/T_{\text{warm}} = \Omega(d^{(L-2)/2})$ up to the gap prefactors.

Proof sketch and provenance. The grind rate d^{L-1} is the online-SGD saddle time for an information-exponent- L target (Ben Arous et al.); the moment-tensor recovery rate $d^{L_\ell/2}$ with prefactor γ_ℓ^{-2} is the sample/iteration cost of estimating a rank-one signal subspace at spectral gap γ_ℓ (Damian et al.; tensor power iteration). Summing over levels and bounding $L_\ell \leq L$ gives $d^{L/2}$; the ratio follows. The statement is asymptotic in d and conditional on the spectral gaps remaining bounded below. $\square$

Proposition 5.2 (Spectral-floor regularisation; finite-size correction). *Let $\hat{\beta}(d_0, d_1) = \log(T_{\text{cold}}/T_{\text{warm}})/\log(d_1/d_0)$ be the empirical saving exponent measured between dimensions $d_0 < d_1$. In the non-asymptotic regime there is an additive floor $T_\bullet = T_\bullet^{\text{asym}} + c_\bullet$ with $c_\bullet = \Theta(1)$ (an $\mathcal{O}(1)$ number of optimiser steps independent of d , e.g. warm-up and detection latency) and a gap-dependent prefactor. Then*

$$\hat{\beta}(d_0, d_1) = \frac{L-2}{2} - \Delta(d_0, d_1; c_\bullet, \gamma_\ell), \quad \Delta > 0, \quad \Delta \rightarrow 0 \text{ as } d_0 \rightarrow \infty,$$

so the measured exponent is biased strictly below the asymptotic $(L-2)/2$ and lies in the observed band ≈ 0.4 – 0.5 for the dimensions tested. This is a finite-size boundary condition, not a failure of the asymptotic growth.

Proof. Write $T_\bullet = A_\bullet d^{\beta_\bullet} + c_\bullet$. Then $\log T_\bullet = \beta_\bullet \log d + \log A_\bullet + \log(1 + c_\bullet A_\bullet^{-1} d^{-\beta_\bullet})$, and the discrete-difference exponent over $[d_0, d_1]$ equals $\beta_\bullet$ minus the positive correction $[\log(1 + c_\bullet A_\bullet^{-1} d_0^{-\beta_\bullet}) - \log(1 + c_\bullet A_\bullet^{-1} d_1^{-\beta_\bullet})]/\log(d_1/d_0)$, which is strictly positive (the floor inflates small- d times more than large- d times) and vanishes as $d_0 \rightarrow \infty$. Applying this to T_{cold} and T_{warm} and subtracting yields the stated $\Delta > 0$ with $\Delta \rightarrow 0$. The numerical experiments, run at $d \in [16, 128]$ with capture detection latency of order tens of steps, sit squarely in the regime where Δ is appreciable, producing the measured 0.35–0.55 slopes while remaining consistent with asymptotic $(L-2)/2$. $\square$

Remark 5.3. We do not claim the asymptotic exponent has been empirically pinned: the direction (saving strictly growing with d) is robust across runs, while the constant $(L-2)/2$ is only consistent with, not confirmed by, the available dimensions. Theorem 5.2 states the obstruction to measuring it precisely rather than asserting the measurement.

Remark 5.4 (The additive-floor structure dictates the correct estimator). The proof of Theorem 5.2 establishes more than a sign: it identifies the bias as the additive $\mathcal{O}(1)$ floor in $T_\bullet = A_\bullet d^{\beta_\bullet} + c_\bullet$. The naive log-log slope $\hat{\beta}(d_0, d_1)$ is biased *because* it ignores $c_\bullet$; the unbiased estimator is the three-parameter nonlinear fit $T(d) = A d^\beta + c$ over a d -range, which is identifiable once the range spans enough decades to separate the power-law term from the constant. We stress the limits of this observation: it is an *estimator-design* correction, not an empirical confirmation. On the tested $d \in [16, 128]$ the three-parameter fit is consistent with $\beta = (L-2)/2$ but the lever arm is too short to pin it (the floor and the power law remain partially confounded at small d). The quantitative open task is unchanged — measure at larger d (or with $c_\bullet$ independently calibrated from the warm-up/detection latency) so that the three-parameter fit becomes well-conditioned. We do not report a recovered exponent here: doing so from synthetic data of the assumed form would recover only what was put in.

6. DISCUSSION: THE SANDBOX BOUNDARY AND THE BRIDGE TO SCALE

The results above are theorems within a controlled class. The class has precise boundaries; we call it the *sandbox boundary*.

(B1) Distributional boundary. Theorem 2.3 requires a symmetric product measure with $\Sigma = I_d$ (Theorem 2.1). Natural data are neither independent across coordinates nor light-tailed nor symmetric. The exact bridge is: replace coordinatewise reflection invariance by an approximate symmetry on a learned whitened frame, and control the resulting bias by the operator norm of the deviation $\mathbb{E}[xx^\top] - I$ together with the magnitude of the leading odd cumulants. The identity $\prod_i \cos \theta_i$ then acquires a perturbative correction of order (third-cumulant) $\times$ (off-diagonal covariance), which must be bounded for heavy-tailed data before the universality transfers. The following makes the correction precise in the *finite-cumulant*, *small-anisotropy* regime.

Proposition 6.1 (First-order correction to the product-cosine law). *Let x have mean zero, covariance Σ with $\|\Sigma - I_d\|_{\text{op}} =: \delta < 1$, and finite third and fourth cumulants, with $\kappa_3 := \max_i |\text{cum}_3(x_i)|$ the leading odd cumulant. Then the leap correlation satisfies*

$$\Gamma = \prod_i \cos \theta_i + R, \quad |R| \leq C \delta \kappa_3 + O(\delta^2 + \kappa_3^2),$$

with C depending only on the leap degree, uniformly over directions $\{\theta_i\}$.

Proof sketch. Expand the leap correlation as a Wick/cumulant series around the symmetric isotropic reference, for which all odd cumulants vanish and $\Sigma = I$ gives exactly $\prod_i \cos \theta_i$ (Theorem 2.3). The first non-vanishing correction is bilinear in the two sources of asymmetry: the off-diagonal/excess covariance $\Sigma - I$ (operator norm δ) and the leading odd cumulant κ_3 (which switches on the odd Hermite channels that reflection symmetry had annihilated). All cross terms are controlled by Hölder against the finite fourth moment; higher orders are $O(\delta^2 + \kappa_3^2)$. $\square$

Remark 6.2 (The truly heavy-tailed case stays open). Theorem 6.1 requires *finite* third and fourth cumulants and small anisotropy $\delta < 1$. It therefore does *not* cover genuinely heavy-tailed data ($p < 2$, infinite variance or non-existent third cumulant), where the expansion has no leading term and the universality may fail outright rather than perturbatively. Establishing (or refuting) a product-cosine law for heavy-tailed x — via a stable-law rather than cumulant expansion — is the open part of B1.

(B2) Target-complexity boundary. The structural targets are low-degree algebraic monomials (finite information exponent, finite leap). Real targets may have unbounded or effectively infinite index. Theorem 3.6 extends Theorem 3.3 from a single leap coordinate to a spectrum of leap exponents *under a spectral-decoupling hypothesis*, where the SNR gate becomes a gate *per spectral band* and the phase diagram is a staircase of thresholds $c_\ell^* = \alpha_\ell / \sqrt{1 + \alpha_\ell^2}$ rather than a single bifurcation.

The bridge that remains open is the *coupled* regime (Theorem 3.7): when bands interact, the captures cease to be independent and the thresholds renormalise sequentially.

(B3) Holonomy boundary. Theorems 4.2 and 4.3 are abelian (linearised, scalar transport): there the Wilson loop is a U(1) phase, the absorption flow is the heat flow on the curvature 2-form, and end-to-end training drives the harmonic holonomy Φ_{hol} to zero (the proven content). Deep vision and language architectures carry non-abelian transport in $\mathcal{O}(n)$ with $n \gg 1$, where the harmonic projection is replaced by the analysis of a flat-connection moduli space and the absorption flow becomes a Yang–Mills-type gradient flow on a principal bundle. We state the expected behaviour as a conjecture and are explicit that it is *not* proved.

Conjecture 6.3 (Flatten-to-class dichotomy, non-abelian holonomy). Let A_t be the end-to-end gradient flow of the Wilson action $\mathcal{E}(A) = \sum_\gamma \|W_\gamma(A) - I\|_F^2$ over connections on a principal $\mathcal{O}(n)$ -bundle ($n \gg 1$). Then (i) $\mathcal{E}(A_t)$ is non-increasing and A_t converges subsequentially to a flat connection A_∞ (zero curvature); but (ii) the flow cannot change the conjugacy class of the holonomy representation $\rho_A : \pi_1 \rightarrow \mathcal{O}(n)$, so residual Wilson loops survive iff ρ_A is topologically non-trivial (non-trivial in $\text{Hom}(\pi_1, \mathcal{O}(n))/\text{conj}$). Absorption thus flattens curvature to the nearest flat connection within the initial class but does not cross between classes.

The abelian reduction of part (i) is Theorem 4.3. The obstruction to a proof at $n \gg 1$ is twofold and genuine: the Wilson action is *non-convex* on the non-abelian moduli space (so monotone descent need not imply convergence to a single limit), and the gradient flow is formally the harmonic-map/Yang–Mills heat flow, which is known to admit *finite-time bubbling* in the relevant dimensions — a singularity that could in principle change the holonomy class and thereby *refute* part (ii). A simulation at small n (e.g. $\mathcal{O}(3)$ on a single cycle with full gauge freedom) cannot discriminate these possibilities: with the holonomy class fixed as a target the flow trivially reaches it, which tests neither non-convexity nor bubbling and is not evidence for the conjecture.

Concrete next step. Prove part (i) rigorously in the *near-flat* (linearised) non-abelian regime, where $\mathcal{E}$ is convex and the flow is the Hodge heat flow on $\mathfrak{o}(n)$ -valued 1-forms: there curvature decays to its harmonic representative and the class is locally preserved. The frontier is then sharply posed: does the *full* (non-linearised) flow at $n \gg 1$ develop bubbling that changes the class? This is the sharpest theoretical question raised by this work, and it remains open.

(B4) Scale and optimiser boundary. The Absorption–Constraint Duality has been verified in shallow networks under plain gradient descent and AdamW. Adaptive optimisers modulate the absorption (they normalise amplitudes, strengthening absorption) and the duality’s quantitative form at large depth and width is untested. The grokking-delay exponent, in particular, was found to be optimiser-dependent, which is consistent with absorption being optimiser-modulated rather than universal. The concrete next step is narrow and checkable: model the AdamW preconditioner as a diagonal amplitude normalisation $w \mapsto w/\sqrt{v}$ in (1) and verify whether it maps the harmonic decay rate $2 \mapsto 2/\sqrt{v}$ (a single modified-OU computation), which would predict the observed optimiser-dependence of the grokking-delay exponent. This is a one-coordinate calculation, not the full deep-width duality, which stays untested.

Within these boundaries the theory is sharp, internally consistent, and falsifiable: it makes the product-state prediction (confirmed across distributions), the SNR gate (confirmed including its α -dependence), and the *abelian* holonomy dichotomy (Theorem 4.3, confirmed on both the frozen and trainable sides); the non-abelian dichotomy (Theorem 6.3) is stated as a conjecture, not confirmed.

Its value is not the claim of universality but the identification of a single mechanism — absorption of anomalies by trainable degrees of freedom, and their reawakening under constraint — that unifies what were previously distinct conjectures and that says, precisely, where each law should and should not hold.

APPENDIX A. SUMMARY OF VERIFIED RATIOS

result	prediction	empirical ratio (estimate/prediction)
Theorem 2.3, Gaussian	$\prod_i \cos \theta_i$	0.999
Theorem 2.3, Uniform/Laplace	$\prod_i \cos \theta_i$	1.001–1.002
Theorem 3.3, threshold at $\alpha = 1$	$c^* = 1/\sqrt{2} \approx 0.707$	gate observed at $c \approx 0.7$
Theorem 3.3, α -scaling	$c^* = \alpha/\sqrt{1 + \alpha^2}$	threshold tracks α
Theorem 4.3(i), frozen	residual = Φ_{hol}	1.000
Theorem 4.3(ii), trainable	$\Phi_{\text{hol}}^{\text{eff}} \rightarrow 0$	loss flat in Θ

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